

Whether government policies
effectively control the epidemic?
Take Taiwan for example.

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Introduction

In this essay, I am going to discuss whether Taiwanese government policies effectively control the epidemic.

When COVID-19 spreads all over the world, the government from every country implements measures such as quarantine, public event cancellation and face covering. Due to cultural differences, different governments can have strict or loose policies. For example, Taiwan government regulated residents to wear face masks imperatively in public areas and a name-based rationing plan when COVID-19 broke out in the early of 2022. On the other hand, the UK government unwound restrictions on wearing masks at the same time despite high confirmed cases.

Compared with other countries' policies, it can be seen that Taiwanese government didn't restrict people's movement too much. They used a soft lockdown policy and detected patients who were infected. They encouraged people to stay at home, and stand together in this difficult time with the government (Chan et al., 2022). In contrast to Vietnam, the reason why they are successful is because their leaders used restrictive policies including movement restrictions, lockdown, contact tracing, quarantine and social distancing (Van Nguyen et al., 2021). Although Vietnamese confirmed cases were relatively lower than other Asian countries, they implemented restrictive measures. As for the Indonesian government, because they have a diversity of institutions which conduct their local policies, it was hard to listen to the government's response to the outbreak. The community's acceptance could be the government's key to combat the virus (Roziqin et al., 2021). On the other hand, European governments are not willing to go against core principles of democracy because decision-makers have to take responsibility for their actions after the pandemic is over (Engler et al., 2021). However, if governments implemented stringent regulations once the pandemic starts, they could save more lives (Fuller et al., 2021).

Because the government policy has influences and real power to restrict people's movement and impose a fine if it is illegal. I am curious about whether confirmed

cases are related to the government policy. My method is to use a dataset from Our World in Data. I also download Policy Responses from the same website in order to combine datasets and to see whether they have a relationship. In order to combat COVID-19, Taiwanese government has tried many methods. The most obvious approach is importing vaccines such as BNT and Moderna, and everyone should get vaccinated at least once in order to reduce the chance of serious symptoms, even death. However, it is not enough with only one vaccination. In the past, wearing a mask was kind of polite to others when people felt ill or had poor skin condition, and now it has become one of the regulations in Taiwan for at least one year.

The reason why I think it is an important topic to this dataset is because the government can check whether their regulations are effective or not. In general, the number of confirmed cases are not only because of government action but also people who are willing to abide by the law. In addition, the number of doctors and hospital beds are also indicators. However, Policies Response in Our World in Data is an easy read and relatively complete data which I think is persuasive for the report. I choose Face Covering Policies, Public Events and Stay-at-Home Restrictions datasets as government policy indicators.

Methodology

1. Explain the datasets

The main dataset I use is the complete COVID-19 dataset from Our World in Data. The date is from 24th February, 2020 to 3rd July, 2022, which depends on when you download the dataset. It mainly includes locations, confirmed cases, death, test, vaccination and some demographic information. According to the topic, I choose location, dates and new cases as my indicators. On the other hand, I also download three different policy datasets from the same website, they are Face Coverings, Cancellation of Public Events and Gatherings and Stay-at-Home Restrictions. The reason why I choose these three policies is because I assume they are main restrictions affecting new cases compared to other datasets in the website. These three policies directly affected the spread of the virus. If I had more time, I would add International and Domestic Travel Restrictions. Taiwanese government tends to use

soft lockdown policy and encourages people to follow regulations (Chan et al., 2022). It can be explained that people are willing to abide by the law to sacrifice self-interest for the good of all. Unlike the Vietnamese government who understood the threat of COVID-19. However, they restricted people's movement compulsively (Van Nguyen et al., 2021). Overall, the three policies I mentioned above are strong to be a study reference.

2. Pre-process and clean the data

First of all, the epidemic in Taiwan was more significant in 2021 and 2022 when looking at the dataset. I want to filter out "Taiwan" and the date was from 01/01/2021. I use the "lubridate" package which is intuitive for dates, and use "vars" to separate dates into three columns such as year, month and day. The reason I separate the date is because the filter does not understand the date format. It may filter out the wrong date range. After filtering out "Taiwan" and the year is equal to 2021 by rows which was learnt from week 4 practical, then I use the select function to select columns of location, date and new cases. This dataset is defined as "covid-data". For the other three policies, I use similar codes, but I use "%>%" to connect the functions. In addition, I remove the first and second columns in order to merge the datasets easily later.

Merging into one dataset is a good method to evaluate. Before I concat these four datasets by date, I find the date column name of covid-data is "date". However, I find the date column name of other datasets is "Day". So I changed the date column name of covid-data to "Day" in order to "left_join" the other three data which I also learnt from week 4 practical. The advantage of "left join" is to connect two datasets according to the left dataset, so I do not need to filter exact dates for matching the same length. Subsequently, I find there are some duplicate columns such as year, month and day, so I use "covid_data[, -c(5,6,7,9,10,11,13,14,15)]" to remove them.

3. Analyse the data in 2021

Reviewing processed covid-data, I find only new_cases is continuous variable, others are categorical variables. There is no suitable analysis method for one continuous variable and one categorical variable. The good way is to replace new_cases with categorical variables and take the Chi-square test which I learnt

from Data Analysis week 6 lecture. I use “cut” to make new_cases break into five sections, and label them from zero to four. This explains the number of new cases that are defined by five different severity levels. Afterwards, I use “cbind” to add a categorical new_cases column in covid_data. Finally, I use “chisq.test” to evaluate the relationship between the number of new cases and policy responses. One is new_cases versus facial_coverings, another is new_cases versus cancel_public_events, and the other is new_cases versus stay_home_requirements.

4. Use another method to process and analyse the data in 2022

In fact, I originally would like to analyse data from 2021 to 2022. But after reading those three policy datasets, I find the variables represent all 0 except face covering in the time during 2022. This cannot be tested because 0 is considered NULL in R. As a result, I only choose the year 2021 to take the Chi-square test.

However, I use another method to continue to conduct my report which is for comparing the relationship between the vaccination rate and the number of confirmed cases and the death rate. Firstly, I again read csv from the original completed dataset and define it as covid_data_2022. Then I separate the date into three columns as the same. At this time, I filter the country as “Taiwan” and the year is “2022”. Differently, I select location, date, people_vaccinated, total_cases and total_deaths as the columns. When I read the dataset, I find some cells are null. I also find these data are actually cumulative which can be used by “zoo” packages in order to replace null with the previous data. But be careful that it should also be set parameter “na.rm = FALSE”, otherwise if there are no earlier non-nulls then the null value will be omitted. I also replace the null value at people_vaccinated[1] with the value on 31/12/2021 which is 18712858. You cannot imagine people vaccinating over 10 million times within a day.

5. Visualise the data

5.1 The result evaluated by 2021

First of all, I import ggplot2 to plot, I also import gridExtra for combining multiple plots on one graph. This technique I learnt from Data Visualisation week 7 practical for continuous data. For the first plot, I use a line chart comparing the number of new cases and face covering policies because a line chart is a good way to observe the

trend. In order to look intuitively, I rescale the number of new cases which are divided by 100. Furthermore, I use two different colours displaying the results. For other indicators, `cancel_public_events` and `stay_home_requirements`, I also use the same method. In the end, I conduct “`grid.arrange`” to link three plots into one straight graph.

5.2 The result evaluated by 2022

For the second processed dataset, I use three bar charts displaying the number of people vaccinated, total cases and total deaths by date. I originally wanted to use a histogram because of cumulative data, but I find that it does not work because I cannot set the date as X-axis. Again, I divide the number by one million in order to make plots more easy to read. The bar colours are customised in light green, light blue and purple. The reason why I chose the bar chart to represent the results is because they display the areas which are easy to look at where the time is gradually growing. When using “`grid.arrange`” to connect three plots, it is convenient for people to compare.

Lastly, although the confirmed cases in 2022 were much higher than in 2021, I can calculate whether the number of deaths is relatively high or not. I use `ggplot2` to plot a line chart, the X-axis is the date, and the Y-axis is `total_deaths` divided by `total_cases`. The result shows that the mortality rate dropped sharply.

Results and discussion

	location	Day	new_cases	facial_coverings	cancel_public_events	stay_home_requirements
1	Taiwan	2021-01-01	3	2	0	0
2	Taiwan	2021-01-02	6	2	0	0
3	Taiwan	2021-01-03	4	2	0	0
4	Taiwan	2021-01-04	3	2	0	0
5	Taiwan	2021-01-05	2	2	0	0
6	Taiwan	2021-01-06	2	2	0	0
7	Taiwan	2021-01-07	3	2	0	0
8	Taiwan	2021-01-08	3	2	0	0
9	Taiwan	2021-01-09	3	2	0	0
10	Taiwan	2021-01-10	0	2	0	0

The graph above shows part of covid_data. Only new_cases is a continuous variable, the others are categorical variables. The variables can be explained from a global panel database of pandemic policies written by Hale et al (2021). I make the table down below.

Policies	Cancel public events	Stay at home	Face coverings
0	No measures	No measures	No policy
1	Recommend cancelling	Recommend not leaving house	Recommended
2	Require cancelling	Require not leaving house with exceptions for daily exercise, grocery shopping, and 'essential' trips	Required in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible
3	x	Require not leaving house with minimal exceptions (e.g. allowed to leave only once every few days, or only one person can leave at a time, etc.)	Required in all shared/public spaces outside the home with other people present or all situations when social distancing not possible
4	x	x	Required outside the home at all times, regardless of location or presence of other people
blank	No data	No data	No data

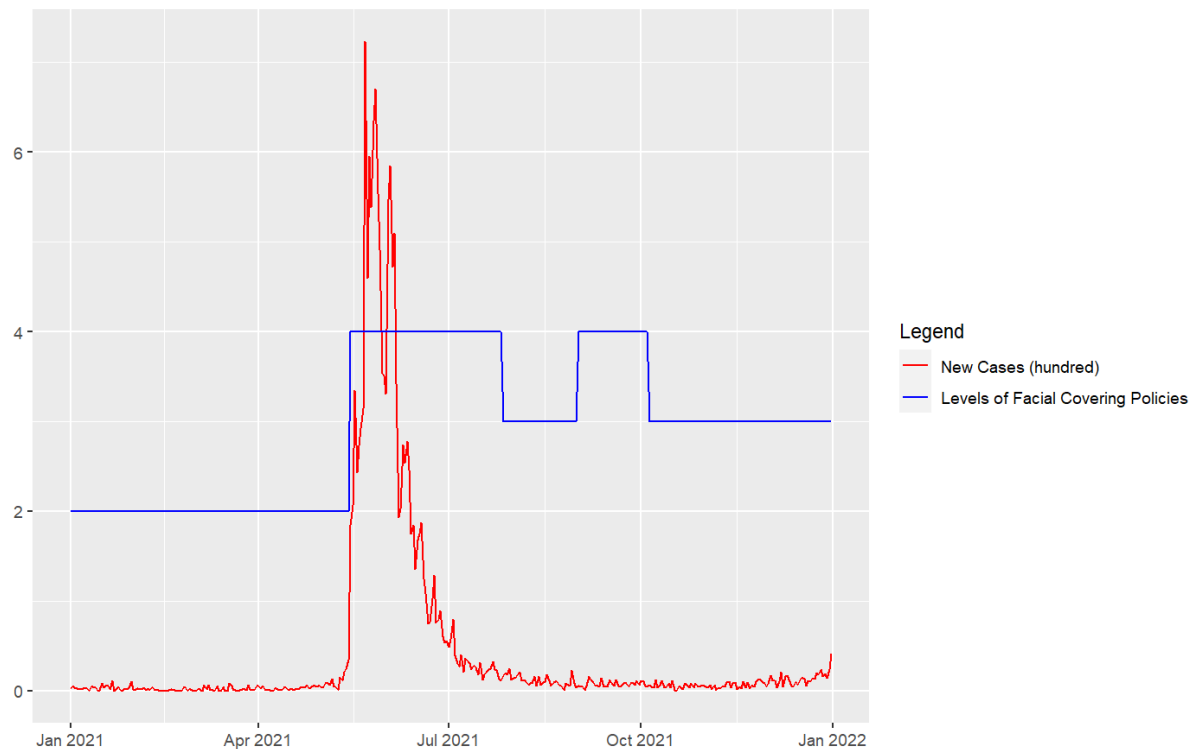
In order to use the Chi-square test, I transfer new_cases into categorical data added behind the last column as level_new_cases which shows below.

	location	Day	new_cases	facial_coverings	cancel_public_events	stay_home_requirements	level_new_cases
1	Taiwan	2021-01-01	3	2	0	0	0
2	Taiwan	2021-01-02	6	2	0	0	0
3	Taiwan	2021-01-03	4	2	0	0	0
4	Taiwan	2021-01-04	3	2	0	0	0
5	Taiwan	2021-01-05	2	2	0	0	0
6	Taiwan	2021-01-06	2	2	0	0	0
7	Taiwan	2021-01-07	3	2	0	0	0
8	Taiwan	2021-01-08	3	2	0	0	0
9	Taiwan	2021-01-09	3	2	0	0	0
10	Taiwan	2021-01-10	0	2	0	0	0

Before conducting the Chi-square test, I state the null hypothesis that there is no association between level_new_cases and other three policies such as face coverings, public events cancellation and stay at home requirements. The X-squared result of the Chi-square test for level_new_cases and facial_coverings is 90.402. Relatively, the P value is smaller than $3.853e-16$. The X-squared for level_new_cases and cancel_public_events is 119.69, and the P value is also smaller than $2.2e-16$. On the other hand, The X-squared for level_new_cases and stay_home_requirements is 119.33, and the P value is smaller than $2.2e-16$. The P value of these three tests are all smaller than 0.01 which means that we can reject the null hypothesis for all three tests. That is, they might have a relationship to confirmed cases.

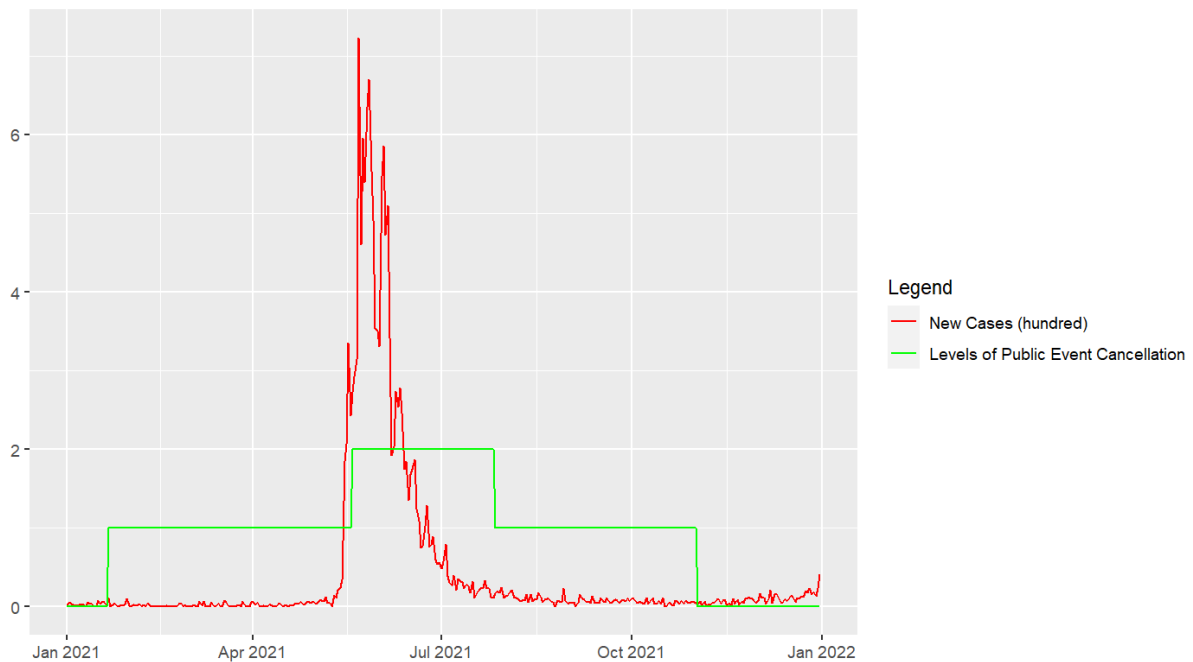
Let's see the first line chart which displays the number of new cases and levels of facial covering policies. This time I do not use level_new_cases because I would like to use the timeline of new cases to see how restrictive for face coverings was. It can be seen that in around May 2021 the new cases increased sharply, and the government raised the restriction from "Required in some specified shared/public spaces outside the home" to "Required outside the home at all times". The new cases increased immediately due to captains of China Airlines cluster infection events (Wikipedia, 2021).

Levels of Facial Covering Policies in 2021



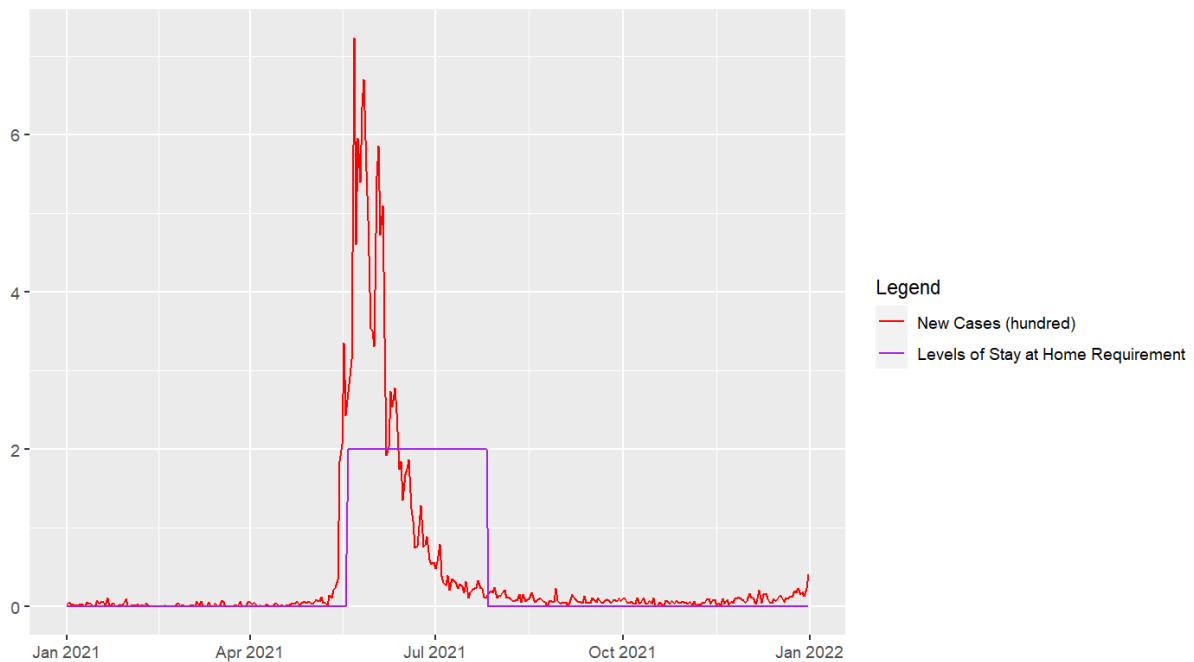
The second line chart displays that the government strictly asked for public events cancellation after May 2021. However, the policy only lasted for 3 months and then turned back to recommend cancelling. After 4 months, there was no measure for public events cancellation. For a democratic country, Engler et al. (2021) note that “domestic factors strongly shape policy making even when faced with a global threat and when problem pressure is high.” When people in the country have more freedom, the government is less likely to limit people’s actions (Engler et al., 2021). The second line chat can be proved that Taiwanese government did not want to prohibit citizens’ activities, but in severe times, it should be recommended or slightly restricted for mutually beneficial symbiosis.

Levels of Public Event Cancellation in 2021

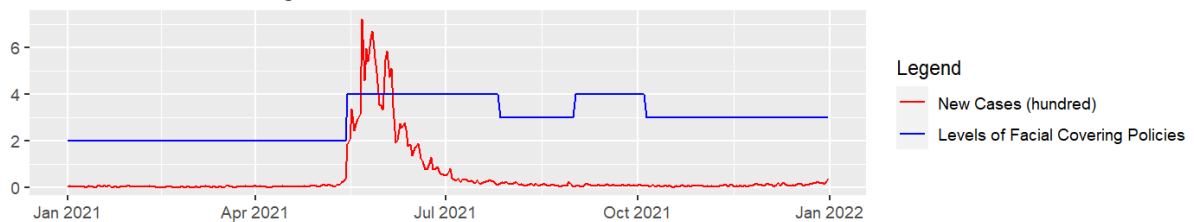


It is restrictive that people did not expect to be asked to stay at home before May 2021, but after the cluster infection events, people are required to not leave home except daily needs. Since infection levels are rising and vaccination numbers remain low, this led to online school courses and working from home, it also rose up online shopping, food delivery and 3C products. Take food delivery to be more specific, although customers could not dine in restaurants, they could rely on food delivery (Robert, K., 2021).

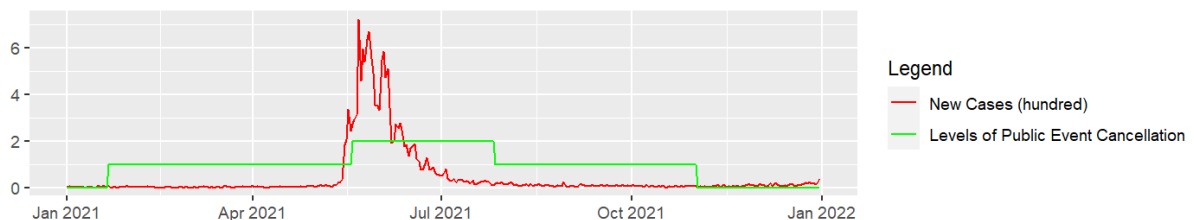
Levels of Stay at Home Requirement in 2021



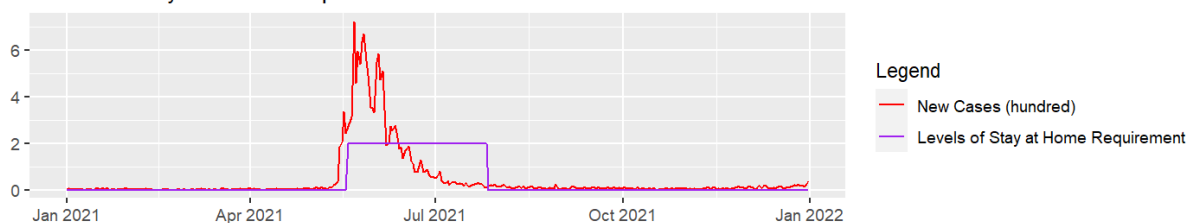
Levels of Facial Covering Policies in 2021



Levels of Public Event Cancellation in 2021

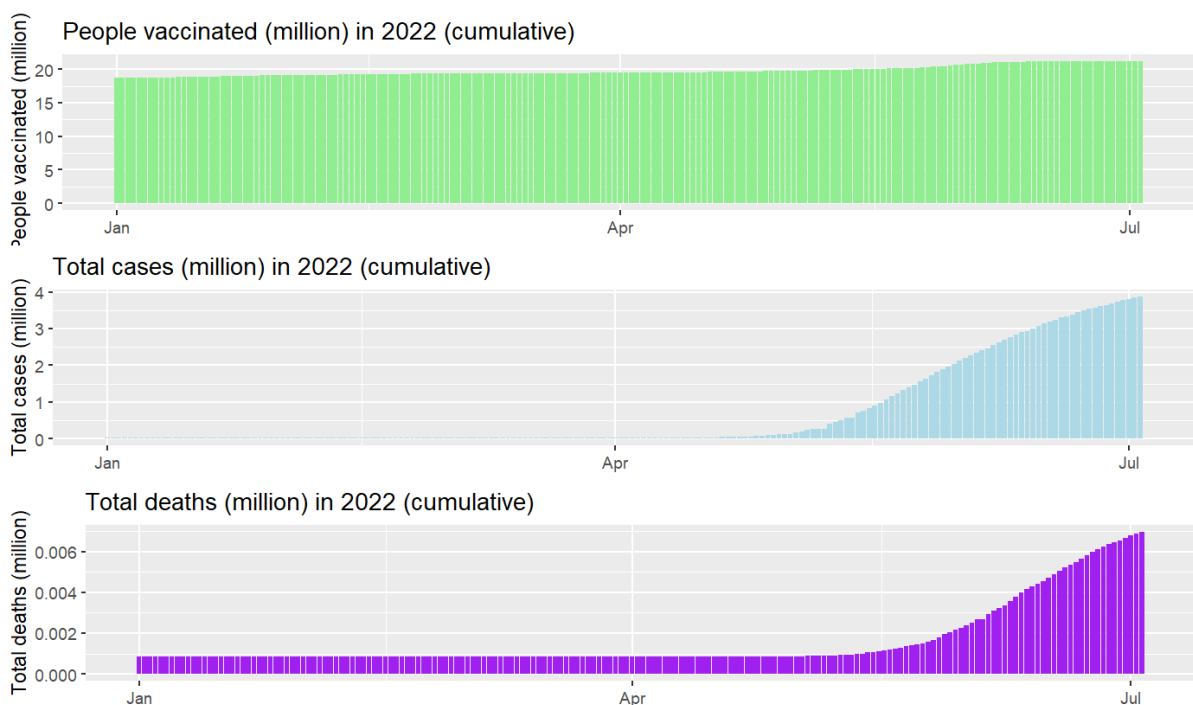


Levels of Stay at Home Requirement in 2021

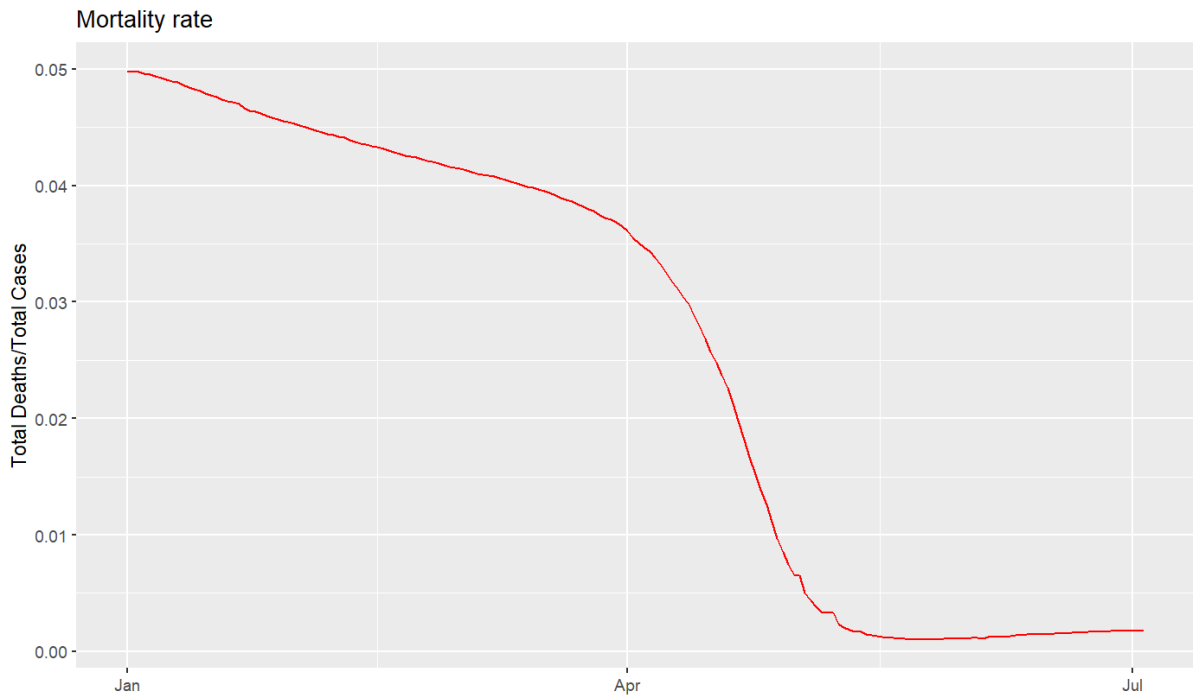


When looking at covid_data_2022 dataset, you can see the variables are all cumulative. It is clear that the number of people vaccinated is much more than in 2021. Some countries have been very generous in donating vaccines to Taiwan since 2021. For example, the Office of the President, Republic of China (Taiwan)

(2021) stated that the United States government has donated a total of 4 million COVID-19 vaccine doses to Taiwan. In the graph below, the number of people vaccinated were around 19 million, however, in July, over 20 million people were vaccinated within 23.6 million citizens, which is over 90% of people who have gotten the first dose of vaccination. Total cases have been substantial growth since around May 2022 which might lead to growth of the number of deaths. When viewing the blue bar chart and the purple bar chart at the same time, you can see that after the growth of total cases, the total deaths gradually increased from 2000 to almost 7000.



Take examples from European countries, it can save more lives if the government implements stringent policies earlier (Fuller et al., 2021). Although Taiwanese government has implemented some measures since 2021, they did not have enough vaccines until the late of 2021. The line chart below displays the mortality rate calculated by total deaths divided by total cases. Despite a high mortality rate in the early of 2022, the rate has sharply declined after April. The rate is lower than 1% consistently after May, which means that the government measures are effective.



Conclusion

Taiwanese government controlled the epidemic once the novel coronavirus broke out in 2021. Although people needed to wear face masks outside at all times since cluster infection events in 2021, mostly people were required to wear masks in some public spaces. The public event cancellation and stay-at-home requirements all raised the level in order to cope with COVID-19 in May 2021; they all soon downgraded the level to normal when the number of new cases decreased to below 100 per day. It can be proved that soft lockdown works on Taiwanese people. Despite null data in public policies in 2022, low mortality rate can also be seen to be successful at epidemic resistance. Due to the donation of vaccines from other countries and private enterprises, the government is able to accelerate distribution of vaccines to the public.

There are some limitations of this study. First of all, governments' regulations depend on their country's culture, it is hard to standardise their level of policies to be more detailed. For example, the north of Taiwan implemented face covering outside due to their local new cases whereas the south of Taiwan did not need to bring masks. But in the dataset, there is no subdivided for regions. We can only see Taiwan's overall policy environment instead of by regions. Secondly, there are less journal articles

which were written about “Taiwan” and “Policies”. Mostly, the articles talk about virus strain and similar kinds of medical research. There is no obvious topic talked about effective government policies.

One of the weaknesses of my analysis is little literature review. If I read more journal articles and online news, I could structure my report better. For example, I can compare other Asian countries near China which were more likely to become high risk areas. Then, I can compare the reason why they were at high risk or not. I can also evaluate their travel and hospitalisations. Afterwards, I am able to use more analysis methods to prove my hypothesis. If I had more time, I would add more graphs in order to make my statement clearly. I would also improve my graph design such as colours, legend, titles.

The takeaway message is that the government can effectively control the epidemic by their policies. According to the Chi-square test, the policies and new cases are significantly related. How restrictive the policies will be depends on different country's cultures. Take Taiwan for example, they did not restrict people for a long time but they still make the epidemic slow down.

R Code

```
Sys.setlocale("LC_ALL", "en_GB.UTF-8") #translate graph into English
```

```
#1. Read data from csv
```

```
covid_data <- read.csv("owid-covid-data.csv", header=TRUE) #Date from 2021/01/16 to 2022/07/03
```

```
facial_coverings <- read.csv("face-covering-policies-covid.csv", header=TRUE) #Date from 2020/01/01 to 2022/12/05
```

```
cancel_public_events <- read.csv("public-events-covid.csv", header=TRUE) #Date from 2020/01/01 to 2022/12/05
```

```
stay_home_requirements <- read.csv("stay-at-home-covid.csv", header=TRUE) #Date from 2020/01/01 to 2022/12/05
```

```
#2. Process with data
```

```
library(tidyverse)
```

```
library(lubridate)
```

```
#Separate date to 3 columns
```

```

covid_data = covid_data %>%
  mutate(date = ymd(date)) %>%
  mutate_at(vars(date), funs(year, month, day))

#Select data
library(dplyr)
covid_data <- filter(covid_data, location=="Taiwan", year == "2021") #Rows
covid_data <- select(covid_data, location, date, new_cases) #Columns

facial_coverings = facial_coverings %>%
  mutate(Day = ymd(Day)) %>%
  mutate_at(vars(Day), funs(year, month, day))
facial_coverings <- filter(facial_coverings, Entity=="Taiwan", year == "2021")
facial_coverings <- facial_coverings[,-1:-2]

cancel_public_events = cancel_public_events %>%
  mutate(Day = ymd(Day)) %>%
  mutate_at(vars(Day), funs(year, month, day))
cancel_public_events <- filter(cancel_public_events, Entity=="Taiwan", year == "2021")
cancel_public_events <- cancel_public_events[,-1:-2]

stay_home_requirements = stay_home_requirements %>%
  mutate(Day = ymd(Day)) %>%
  mutate_at(vars(Day), funs(year, month, day))
stay_home_requirements <- filter(stay_home_requirements, Entity=="Taiwan", year ==
"2021")
stay_home_requirements <- stay_home_requirements[,-1:-2]

```

#3. Combine & select data by "Day"

```

colnames(covid_data)[2] <- "Day"
covid_data = covid_data %>% left_join(facial_coverings, by="Day")
covid_data = covid_data %>% left_join(cancel_public_events, by="Day")
covid_data = covid_data %>% left_join(stay_home_requirements, by="Day")

covid_data <- covid_data[,-c(5,6,7,9,10,11,13,14,15)]

```

```

head(covid_data)
tail(covid_data)

```

#4. Transfer continuous variable (new_cases) to categorical data and add behind the dataset

```

level_new_cases <- cut(covid_data$new_cases, breaks = 5, labels = 0:4)
covid_data <- cbind(covid_data, level_new_cases)

```

```

head(covid_data)
tail(covid_data)

```

#5. Chi-square test analysis for three policies

```
chisq.test(covid_data$level_new_cases, covid_data$facial_coverings, correct=FALSE)
```

```
#X-squared = 90.402, df = 8, p-value = 3.853e-16
```

```
chisq.test(covid_data$level_new_cases, covid_data$cancel_public_events, correct=FALSE)
```

```
#X-squared = 119.69, df = 8, p-value < 2.2e-16
```

```
chisq.test(covid_data$level_new_cases, covid_data$stay_home_requirements,  
correct=FALSE)
```

```
#X-squared = 119.33, df = 4, p-value < 2.2e-16
```

#6. Visualisation

```
library(ggplot2)
```

```
library(gridExtra)
```

```
facial.plot <- ggplot(covid_data, aes(x=Day))+  
  geom_line(aes(y=new_cases/100, col="New Cases (hundred)"))+  
  geom_line(aes(y=facial_coverings, col="Levels of Facial Covering Policies"))+  
  labs(x=NULL, y=NULL,  
        title="Levels of Facial Covering Policies in 2021")+  
  scale_colour_manual(name="Legend",  
                      values=c("New Cases (hundred)"="red",  
                               "Levels of Facial Covering Policies"="blue"))  
plot(facial.plot)
```

```
public.plot <- ggplot(covid_data, aes(x=Day))+  
  geom_line(aes(y=new_cases/100, col="New Cases (hundred)"))+  
  geom_line(aes(y=cancel_public_events, col="Levels of Public Event Cancellation"))+  
  labs(x=NULL, y=NULL,  
        title="Levels of Public Event Cancellation in 2021")+  
  scale_colour_manual(name="Legend",  
                      values=c("New Cases (hundred)"="red",  
                               "Levels of Public Event Cancellation"="green"))  
plot(public.plot)
```

```
home.plot <- ggplot(covid_data, aes(x=Day))+  
  geom_line(aes(y=new_cases/100, col="New Cases (hundred)"))+  
  geom_line(aes(y=stay_home_requirements, col="Levels of Stay at Home Requirement"))+  
  labs(x=NULL, y=NULL,  
        title="Levels of Stay at Home Requirement in 2021")+  
  scale_colour_manual(name="Legend",  
                      values=c("New Cases (hundred)"="red",  
                               "Levels of Stay at Home Requirement"="purple"))  
plot(home.plot)
```

```
library(gridExtra)
```

```
grid.arrange(facial.plot, public.plot, home.plot, ncol=1)
```


#7. Obtain data again & Choose other indicators

```
covid_data_2022 <- read.csv("owid-covid-data.csv", header=TRUE)
```

```
covid_data_2022 = covid_data_2022 %>%  
  mutate(date = ymd(date)) %>%  
  mutate_at(vars(date), funs(year, month, day))
```

```
covid_data_2022 <- filter(covid_data_2022, location == "Taiwan", year == "2022")  
covid_data_2022 <- select(covid_data_2022, location, date, people_vaccinated, total_cases,  
total_deaths)
```

#8. Process with missing value

```
install.packages("zoo")  
library("zoo")  
covid_data_2022 <- na.locf(covid_data_2022, na.rm = FALSE)  
#na.rm = FALSE, in order to keep Null value  
covid_data_2022$people_vaccinated[1] <- 18712858
```

#9. Visualisation

```
people_vaccinated.bar <- ggplot(covid_data_2022, aes(date,  
people_vaccinated/1000000))+geom_bar(stat="identity", fill="lightgreen")+  
  labs(x=NULL, y="People vaccinated (million)",  
    title="People vaccinated (million) in 2022 (cumulative)")
```

```
total_cases.bar <- ggplot(covid_data_2022, aes(date,  
total_cases/1000000))+geom_bar(stat="identity", fill="lightblue")+  
  labs(x=NULL, y="Total cases (million)",  
    title="Total cases (million) in 2022 (cumulative)")
```

```
total_deaths.bar <- ggplot(covid_data_2022, aes(date,  
total_deaths/1000000))+geom_bar(stat="identity", fill="purple")+  
  labs(x=NULL, y="Total deaths (million)",  
    title="Total deaths (million) in 2022 (cumulative)")
```

```
grid.arrange(people_vaccinated.bar, total_cases.bar, total_deaths.bar, ncol=1)
```

```
ggplot(covid_data_2022, aes(date, total_deaths/total_cases))+ geom_line(colour="red")+  
  labs(x=NULL, y="Total Deaths/Total Cases",  
    title="Mortality rate")
```

Reference

Chiu, Y. J., Chiang, J. H., Fu, C. W., Hour, M. J., Ha, H. A., Kuo, S. C., ... & Tsai, F. J. (2021). Analysis of COVID-19 prevention and treatment in Taiwan. *BioMedicine*, 11(1), 1.

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